Interfacing LED/LEDS:

1. Write an ECP to find the second highest digit in a given integer and display its binary equivalent on 8-LEDS (4-Active High LEDS& 4-Active Low LEDS).

2. Write an ECP to display binary equivalent of prime numbers within the range of 0-50 @ 5 seconds. (Note: Use IsPrime() function)

3. Write an ECP to print the binary equivalent of odd digits in a given integer on 8 LEDS @ 5 seconds.

4. Write an ECP to print the binary equivalent of sum of the digits for the given integer on LEDS (4-Active High LEDS& 4-Active Low LEDS).

5. Write an ECP to print the binary equivalent of product of the digits for the given integer on LEDS (4-Active High LEDS& 4-Active Low LEDS).

Note: if the product is >255, Switch on alternate LEDS.

6. Write an ECP to print the binary equivalent of power of 2 numbers with in the range of 0-255 @ 5 seconds.

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Seven Segment Display + Switch:

1. Write an ECP for Single segment only should be lit (starting from segment A) cyclically around the periphery of single seven segment (5 times clock wise and 5 times anti-clock wise @ proper visualization).

2. Write an ECP to print the switch press count within 500 milli seconds on two multiplexed seven segment displays.

Note: Time slice should start after the first switch press

Explanation: Initially 0 will display on two multiplexed displays. Once switch is pressed, then time slice will start. If switch is pressed with in time slice again then time slice starts from 0. If there is no switch press within time slice then display the updated count on two multiplexed seven segment displays. After switch is pressed, within time slice no need to display any value on segments. (This logic is giving support for SMS keypad program implementation)

3. Produce the display on seven segments using switches connected to any port in the pattern explained:

a. First two Segments should be OFF
b. For every SW1 press the value should be incremented and displayed on the seven segments.
c. For every SW2 press the value should be decremented and displayed on the seven segments.
d. For every SW3 press the value should be reset.

4. Write an ECP to display the prime numbers within 0-99 on two multiplexed seven segment displays. (Note: must use IsPrime() function)

5. Write an ECP to display the Fibonacci series numbers (0-99) on two multiplexed seven segment displays.

6. Write an ECP to display the 10 random numbers within 0-99 on two multiplexed seven segment display with respect to 5 seconds.

7. Write an ECP to implement electronic dice using one switch and single segment display.

8. Write an ECP to display 10 random numbers within 0-9 and display the sum of all random numbers on two multiplexed seven segment displays.

Note: display random numbers also.

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Interfacing 16X2 LCD:

1. Write an ECP to take 20 numbers randomly in an array and find the prime numbers in the list of numbers to display on LCD.

Note: must use rand() function

2. Write an ECP to take ten 3-digit numbers randomly in an array and find palindrome numbers in the list of numbers to display on LCD.

Note: must use rand() function

3. Write an ECP to check whether given string is palindrome or not and display result on LCD.

4. Write a program to display the message "VECTOR" on the first Line and "Institute" on the second line of a 2x16 LCD. Then make "Institute" scroll from right to left on second line of LCD screen.

5. Write an ECP to convert first character of each word in to upper case and display the final result on LCD.

Note: Connect two switches. Based on first switch press, take "vector india" as input and do the operation; Based on second switch press, take "vector Hyderabad" as input and do the operation. In both cases print input and output on lcd screen.

Output:

When Switch1 is pressed, display "vector india" on the 1st line of LCD and when the switch1 is released display "Vector India" on the 2nd line of LCD.

When Switch2 is pressed, display "vector hyderabad" on the 1st line of LCD and when the switch2 is released display "Vector Hyderabad" on the 2nd line of LCD.

If none of the switch is pressed, display "waiting for input" on the 1st line of LCD.

6. Write an ECP to display the ASCII table information for A-Z, a-z & 0-9 on LCD screen with respect to 5 seconds.

For example,

1st line: A D H O

2nd line: A 65 41 101

After 5 secs

1st line: A D H O

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2nd line: B 66 42 102 and so on ….

7. Write an ECP to display the binary value for A-Z, a-z & 0-9 on LCD screen with respect to 5 seconds.

For example,

1st line: A 2nd line: 01000001

After 5 secs

1st line: B 2nd line: 01000010 and so on ….

8. Write an ECP to convert first and last character of each word in to upper case and display the final result on LCD.

Note: Connect two switches. Based on first switch press, take "vector india" as input and do the operation; Based on second switch press, take "vector Hyderabad" as input and do the operation. In both cases print input and output on lcd screen.

Output:

When Switch1 is pressed, display "vector india" on the 1st line of LCD and when the switch1 is released display "VectoR IndiA" on the 2nd line of LCD.

When Switch2 is pressed, display "vector hyderabad" on the 1st line of LCD and when the switch2 is released display "VectoR HyderabaD" on the 2nd line of LCD.

If none of the switch is pressed, display "waiting for input" on the 1st line of LCD.

9. Write an ECP to remove extra spaces in a given string and display the final result on LCD.

Note: Connect two switches. Based on first switch press, take " An Apple a day" as input and do the operation; Based on second switch press, take " vector stu !" as input and do the operation. In both cases print input and output on lcd screen.

Output:

When Switch1 is pressed, display " An Apple a day" on the 1st line of LCD and when the switch1 is released display "An Apple a day" on the 2nd line of LCD.

When Switch2 is pressed, display " vector stu !" on the 1st line of LCD and when the switch2 is released display "vector stu !" on the 2nd line of LCD.

If none of the switch is pressed, display "waiting for input" on the 1st line of LCD.

10. Write an ECP to develop any simple game on LCD by using customised characters.

Note: if required you can take any input devices (switch, keypad) support

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Assignments on 4x4 matrix keypad with output devices:

1. Write an ECP to design a menu-based calculator using 4x4 matrix keypad and 16X2 lcd.

Menu: 1. Addition 2. Subtraction …. etc

2. Write an ECP to check whether given number is prime or not and display result on LCD.

Note: use keypad to take the input (multi digit) and must use IsPrime() & ReadNum() functions

3. Write an ECP to take one number (multi digit) from keypad and display its binary equivalent on LCD.

Note: must use BinLCD() function

4. Write an ECP to check whether given number is palindrome or not and display result on LCD.

Note: use keypad to take the input and must use IsPalindrome() function

VECTOR INDIA

Assignments on UART:

1. Write an ECP to accept valid password from user to access security-based devices using UART with 9600 baud rate.

Note: password should contain,

At least one numeric digit, at least one Lowercase Letter, at least one Uppercase Letter, at least one special letter, must not have space (‘ ‘), must not have slash (‘/’) and at least 8 characters.

If someone inputs an invalid password, the code prints: “Invalid Password” on hyper terminal otherwise, it prints: “Valid Password”.

2. Write an ECP to display the below mentioned patterns on hyper terminal with the help of UART0 with 19200 baud rates.

